EQUIPMENT DATA SHEET

E-301 -- Shell and Tube Heat Exchanger

Crude Preheat Service -- Unit 300

Document Number:	DS-E301-001 Rev. 3
Manufacturer:	Alfa Laval (formerly Aalborg Industries)
PO Number:	PM-PO-2016-03045
Year Manufactured:	2016
Year Installed:	March 2017
TEMA Type Designation:	BEM (Bonnet, Single Pass, Fixed Tubesheet)

1. Design Data -- Shell Side

Fluid:	Hot product return (heavy naphtha)
Flow Rate:	850,000 lb/hr
Inlet Temperature:	420 degF (215.6 degC)
Outlet Temperature:	290 degF (143.3 degC)
Operating Pressure:	185 psig
Design Pressure:	300 psig
Design Temperature:	500 degF (260 degC)
Material (Shell):	SA-516 Gr. 70, normalized
Shell ID:	42 inches
Corrosion Allowance:	0.125 inches
Shell Thickness (nominal):	0.750 inches
Baffle Type:	Single segmental, 25% cut
Baffle Spacing:	12 inches (central), 9 inches (inlet/outlet)
MDMT:	-20 degF (impact tested)

2. Design Data -- Tube Side

Fluid:	Crude oil feed (desalted)
Flow Rate:	1,200,000 lb/hr
Inlet Temperature:	250 degF (121.1 degC)
Outlet Temperature:	370 degF (187.8 degC)
Operating Pressure:	195 psig
Design Pressure:	250 psig
Design Temperature:	450 degF (232.2 degC)
Number of Tubes:	450
Tube OD:	0.750 inches (3/4")
Tube BWG:	16 (wall thickness 0.065 inches)
Tube Length (effective):	20 feet
Tube Pitch:	1.0 inch, triangular (30 deg)
Tube Material:	SA-179 (seamless carbon steel)
Tube-to-Tubesheet Joint:	Expanded and seal welded
Number of Passes:	1 (single pass)

3. Thermal Performance Data

The following thermal performance parameters are based on the original design calculations and the as-built guarantee test performed during commissioning in April 2017.

Parameter	Design	Guarantee	Unit
Heat Duty	52.4	51.8	MMBTU/hr
Overall U (clean)	128	125	BTU/hr-ft2-F
Overall U (fouled)	95	--	BTU/hr-ft2-F
LMTD (corrected)	48.2	47.5	degF
Effective Area	7,069	7,069	ft2
Fouling Factor (shell)	0.001	--	hr-ft2-F/BTU
Fouling Factor (tube)	0.002	--	hr-ft2-F/BTU
Shell-side DP (clean)	3.5	3.8	psi
Tube-side DP (clean)	5.2	5.5	psi

4. Current Operating Performance (as of March 2025)

Based on the most recent performance test conducted on March 12, 2025 (ref: WO-5008), the following operating performance was recorded. Note that 31 tubes have been plugged as of January 2025 (ref: WO-5002), reducing the effective tube count to 419.

Parameter	Current	Design (clean)	Status
Overall U (operating)	79	128	DEGRADED
Heat Duty (actual)	45.2	52.4	87% of design
Crude Outlet Temp	348 degF	370 degF	22 degF below
Shell-side DP	11.7 psi	3.5 psi	FOULED
Tube-side DP	6.8 psi	5.2 psi	MARGINAL
Plugged Tubes	31 / 450	0	6.9%

Analysis: The 38% decline in overall heat transfer coefficient from clean design value (128 to 79 BTU/hr-ft2-F) is primarily attributable to shell-side fouling, as evidenced by the 234% increase in shell-side pressure drop (3.5 to 11.7 psi). The crude outlet temperature of 348 degF (versus design 370 degF) means the downstream fired heater (H-301) must compensate with an additional 7.2 MMBTU/hr of fuel firing, at an estimated cost of $12,800/day in additional fuel gas consumption.

Recommendation: Chemical cleaning of the shell side during the Q2 2025 turnaround is expected to restore the overall U to approximately 110-115 BTU/hr-ft2-F (85-90% of clean value). Long-term, the engineering team should evaluate the installation of an automatic backwash system or continuous antifouling treatment (e.g., Nalco EC1270A) to extend the cleaning interval.

5. Nozzle Schedule

Nozzle	Service	Size	Rating	Facing	Material
N1	Shell Inlet	16"	300#	RF	SA-105
N2	Shell Outlet	16"	300#	RF	SA-105
N3	Tube Inlet	18"	150#	RF	SA-105
N4	Tube Outlet	18"	150#	RF	SA-105
N5	Shell Vent	2"	300#	RF	SA-105
N6	Shell Drain	2"	300#	RF	SA-105
N7	Tube Vent	2"	150#	RF	SA-105
N8	Tube Drain	2"	150#	RF	SA-105
N9	Shell DP Conn.	1"	300#	NPT	SA-105
N10	Tube DP Conn.	1"	150#	NPT	SA-105

6. Maintenance History Summary

The following is a summary of significant maintenance events for E-301 since installation:

March 2017: Commissioning and guarantee test. Performance verified within 2% of design.

September 2018: First shell-side chemical cleaning (Zyme-Flow product). U-value restored to 122 BTU/hr-ft2-F.

March 2020: ECT inspection during turnaround. 8 tubes plugged (>40% wall loss). Performance acceptable.

October 2021: Shell-side chemical cleaning. U-value restored to 118 BTU/hr-ft2-F.

April 2023: ECT inspection. 11 additional tubes plugged (total 19). Fouling factor trending higher -- cleaning interval reduced from 24 months to 18 months.

January 2025: ECT inspection (ref: WO-5002). 12 additional tubes plugged (total 31). Performance significantly degraded. Chemical cleaning scheduled for Q2 2025 turnaround.

April 2025: Shell-side chemical cleaning performed (ref: WO-5012). Post-cleaning performance test (ref: WO-5017) confirmed U-value restored to 112 BTU/hr-ft2-F.

7. Interconnected Equipment

E-301 is part of the crude preheat train and is directly interconnected with the following equipment:

Upstream (Tube Side Feed): P-101 (Centrifugal Feed Pump) supplies crude oil to E-301 tube side via 18" line 100-CR-018-A5. Any capacity changes to P-101 directly affect E-301 tube-side velocity and fouling tendency.

Downstream (Tube Side): V-201 (Flash Separator Vessel) receives the heated crude from E-301. The crude outlet temperature from E-301 directly affects V-201 flash efficiency.

Upstream (Shell Side): The hot product return stream from the atmospheric column overhead condenser feeds the E-301 shell side.

Downstream (Shell Side): The cooled product exits E-301 and flows to TK-501 (Storage Tank) via the product rundown cooler.

Note: P-102 (Booster Pump) operates in series with P-101 and indirectly affects E-301 through its impact on total system flow rate and pressure.